

SUMMARY

Results-driven Senior Software Engineer with 5+ years architecting enterprise-grade distributed systems across fintech and education sectors. Expert in Java, Spring Boot, Python, and React, delivering high-performance microservices and AI-powered solutions on AWS/GCP. Proven track record reducing latency by 40%, cutting deployment times by 70%, and improving system reliability through RESTful/GraphQL APIs, Kafka-based data pipelines, and SQL/NoSQL optimization. Strong advocate for security (OAuth, JWT, MFA), TDD, and mentorship in agile environments.

SKILLS

- **Languages:** Java (8–17), Python, JavaScript, TypeScript
- **Frontend Development:** React.js, Redux, Angular, D3.js, Material-UI, HTML5, CSS3
- **Backend Development:** Spring Boot, Spring MVC, Hibernate, Node.js, Flask, Django, GraphQL, RESTful APIs
- **Database Management:** PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch
- **Cloud & DevOps:**
 - AWS: EC2, S3, RDS, Lambda, IAM, KMS, CloudWatch, SIEM
 - GCP: Cloud Functions, Pub/Sub, Compute Engine
 - Docker, Kubernetes, Jenkins, GitLab CI/CD, Terraform
- **Architecture & Integration:** Microservices, RESTful APIs, GraphQL, API Integrations, Swagger, Postman
- **Message Queuing:** Apache Kafka, RabbitMQ
- **Big Data & Streaming:** Apache Kafka, Apache Spark, ETL Pipelines
- **Security & Compliance:** OAuth 2.0, JWT, Multi-Factor Authentication (MFA), Role-Based Access Control (RBAC), PCI-DSS, GDPR
- **Monitoring & Logging:** ELK Stack (Elasticsearch, Logstash, Kibana), CloudWatch, Prometheus, Grafana
- **Caching Strategies:** Redis, Memcached, Application-level Caching
- **Testing & Quality:** JUnit, TestNG, Jest, Selenium, Test-Driven Development (TDD)
- **AI & Machine Learning:** PyTorch, OpenAI API, LangChain, RAG Architecture
- **Tools & Methodologies:** Git, GitHub, GitLab, Maven, Jira, Agile/Scrum, SDLC

WORK EXPERIENCE**Western Union | Senior Software Developer****Feb 2025 – Jan 2026**

- Architected and deployed **scalable microservices** using **Java**, **Spring Boot**, and **Python Flask** with **RESTful** and **GraphQL APIs**, reducing transaction processing latency by **40%** and API response times by **30%** while maintaining **99% system uptime**.
- Engineered **real-time data streaming** solutions with **Apache Kafka** and **Apache Spark**, processing **high-volume financial transactions** and enabling seamless integration with **AI-driven fraud detection** systems, improving detection accuracy by **25%**.
- Optimized **PostgreSQL database** performance by **50%** through advanced **indexing strategies**, **query optimization**, and **partitioning** for high-frequency transactions, while designing robust schemas ensuring data integrity and **PCI-DSS compliance**.
- Developed **interactive fraud analytics dashboards** using **React.js** and **D3.js**, empowering risk analysts with **real-time insights** and data-driven decision-making capabilities for pattern detection and anomaly identification.
- Implemented **enterprise-grade security frameworks** including **OAuth 2.0**, **JWT authentication**, and **multi-factor authentication (MFA)**, ensuring **PCI-DSS**, **GDPR**, **FINRA**, and **SEC compliance** while reducing audit risks by **20%**.
- Automated **CI/CD pipelines** using **GitLab**, **Docker**, and **Kubernetes**, reducing deployment times by **70%** and integrating **automated testing**, **security scanning**, and **rolling updates** for zero-downtime releases.
- Designed and deployed scalable **AWS cloud infrastructure (EC2, S3, Lambda, RDS)** adhering to **FFIEC** and **SOX compliance** standards, achieving **99% uptime** for mission-critical banking applications.
- Established comprehensive **testing strategies** achieving **95% code coverage** through **unit** and **integration testing** with **JUnit** and **TestNG**, minimizing post-deployment defects and ensuring software quality.
- Mentored engineering teams on **Spring framework**, **microservices architecture**, **RESTful design patterns**, and troubleshooting methodologies, enhancing team technical proficiency and delivery efficiency.
- Integrated **third-party payment provider APIs** with **rate limiting** and **retry mechanisms** using **Spring Boot** and **Redis caching**, enabling seamless transaction routing across multiple providers and reducing integration failures by **45%**.
- Implemented **distributed logging** with **ELK Stack** and **CloudWatch**, creating centralized log aggregation across **15+ microservices**, reducing **mean-time-to-resolution** for production incidents by **60%** through enhanced debugging and performance monitoring.

Software Engineer(Research Assistant) | California State University, Long Beach**July 2023 – Dec 2024**

- Architected **microservices-based scheduling platform** using **Java 17**, **Spring Boot**, and **Node.js**, managing **50,000+ students** across **2,500+ sections** with **99.8% uptime** during peak registration.
- Engineered intelligent **course scheduling engine** with **conflict detection algorithms**, automatically identifying time conflicts and prerequisite violations, reducing manual registration errors by **85%**.
- Developed **real-time seat availability dashboard** using **React.js**, **Redux**, and **WebSocket**, providing instant course opening updates and reducing student support tickets by **60%**.
- Optimized **PostgreSQL database** schema for academic calendars and room assignments, improving query performance by **70%** through **indexing**, **materialized views**, and **query optimization**.

- Implemented **Elasticsearch-powered course search** with advanced filtering capabilities, achieving **sub-80ms response times** and improving discovery experience for **50K students** across **300+ programs**.
- Built **automated waitlist management system** with **priority queue algorithms**, processing **5,000+ movements** per cycle and ensuring fair seat allocation without manual intervention.
- Deployed **Redis caching layer** for course catalogs and student schedules, reducing database load by **55%** and improving page load times from **3.2s to 0.8s**.
- Achieved **90% code coverage** through **unit and integration testing** with **JUnit, Jest, and Selenium**, catching **95%** of defects before production deployment.
- Built **AI-powered course advisor** using **RAG architecture** with **OpenAI embeddings** and **vector database**, enabling students to query course information conversationally, answering **5K+ During registrations** with **92% accuracy** on prerequisites.
- Engineered **intelligent course recommendation system** leveraging **RAG pipeline** with **LangChain** and **sentence transformers**, analyzing student transcripts and degree requirements to suggest optimal course sequences, improving degree planning for **5,000+ students**.

Founding Software Engineer | Zillion Technologies (SAIX Inc.)

July 2021 – Jan 2023

- Architected and deployed **AI-driven identity threat detection platform** using **Python, TensorFlow, Spring Boot microservices**, and **RESTful APIs**, processing **10M+ identity events** daily to detect vulnerable identities and prevent breaches, reducing insider threat detection time by **80%**.
- Engineered **real-time anomaly detection engines** using **machine learning algorithms (Random Forest, SVM, CNN)** and behavioral analysis models, analyzing user access patterns and privilege escalations with **95% detection accuracy**, identifying zero-day identity vulnerabilities before exploitation.
- Built **high-throughput data ingestion pipelines** using **Apache Kafka** and **Python Flask**, streaming access logs from **SIEM, IAM, and Active Directory** systems, processing **50M+ security events** daily to establish ground truth and identify control failures.
- Developed **automated remediation orchestration system** using **Python** and **RESTful APIs**, triggering intelligent workflows for access revocation and privilege de-escalation, reducing data exfiltration risks by **90%** and automating **85%** of remediation tasks.
- Designed and developed **interactive threat visualization dashboards** using **React 16.8, D3.js, and Material-UI**, enabling security analysts to monitor identity risks and attack patterns in real-time across **50K+ enterprise identities** and connected systems.
- Engineered **cloud-native security platform** on **Google Cloud Platform** using **Cloud Functions, Pub/Sub, and Terraform**, implementing **serverless architecture** for real-time identity risk assessment, reducing operational costs by **40%** and enabling elastic scalability.
- Established **CI/CD pipelines** using **Jenkins** and **Docker**, reducing model deployment time by **60%**, implementing **automated security scanning** and gradual rollout strategies for ML-based threat detection updates with **99.5% uptime**.
- Built **multi-tenant identity platform** with **tenant data isolation** using **Spring Boot** and **PostgreSQL row-level security**, supporting **50+ enterprise clients** with **99.9% uptime** and implementing strict data segregation preventing cross-tenant data leakage.

Junior Software Engineer | Magnitude Software

Dec 2020– July 2021

- Developed **enterprise IAM applications** using **Java 8, Spring Framework, Hibernate ORM, and RESTful APIs**, building user provisioning and **role-based access control (RBAC)** for **50K+ enterprise users** with **99.5% uptime**.
- Built **database-driven identity management solutions** with **SQL Server**, designing optimized schemas for user profiles and access certifications, implementing **stored procedures** that reduced query execution time by **40%**.
- Contributed to **AWS cloud migration** of legacy identity systems, refactoring authentication services for **EC2** and **RDS** deployment, implementing **IAM roles** and **KMS encryption** for secure credential management.
- Built **automated user provisioning workflows** using **Spring Boot** and **Java scheduled tasks**, orchestrating employee onboarding and offboarding across **15+ enterprise applications**, reducing manual account creation time from **4 hours to 15 minutes**.
- Developed **quarterly access review campaigns** using **Java Streams** and **database queries**, automating manager approval workflows for **50K users**, improving compliance audit readiness and reducing certification cycle time by **70%**.

EDUCATION

California State University, Long Beach – Long Beach, CA | Master's in Computer Science

Jan 2023 – Dec 2024

CERTIFICATIONS

Certified API Security Analyst – APIsec University